

Round 1 Interview (Deadline: 36 hrs)

Welcome to Round 1 of the Assignment Round

The current assignment consists of the following task:

Technical & Development Task

You are requested to carefully read through the provided document and watch the video instructions (applicable to this task) in full before you begin implementing your solution.

Once you have completed the task, please use the submission URL provided at the end of the document (Google Form link) and fill in all the required fields.

Task: AI-First CRM HCP Module – Log Interaction Screen (Technical)

Note: Use of LangGraph and an LLM is mandatory. Submissions that do not incorporate these will not be accepted.

Objective: Conceptualize and outline the key components of an AI-first Customer Relationship Management (CRM) system, specifically focusing on the Healthcare Professional (HCP) module. You should approach this task with the mindset of a life science expert designing a tool for field representatives.

Core Requirements & Tech Stack:

- **Functionality:** Design the "Log Interaction Screen." This screen must offer users the flexibility to log interactions with HCPs via either a structured form or a conversational chat interface.
- **Frontend:** **React** UI with **Redux** for state management.
- **Backend:** **Python** with **FastAPI**.
- **AI Agent Framework:** **LangGraph**.
- **LLMs:** Utilize Groq(<https://console.groq.com/docs/models>), specifically **the gemma2-9b-it model** (assume you will create a new API token for this). You may also consider **llama-3.3-70b-versatile** for context.
- **Database:** MySQL/Postgres SQL
- **Font:** Google Inter.
- **You may use Gemini 2.5 Pro (1-month free trial) or ChatGPT 5.0 for all coding and research, with zero human-written code.**

- Video URL: 📺 Instructions(Task 1).mp4

Log HCP Interaction

Interaction Details

HCP Name

Interaction Type Meeting

Date

Time

Attendees

Topics Discussed

Summarize from Voice Note (Requires Consent)

Materials Shared / Samples Distributed

Materials Shared

No materials added.

Samples Distributed

No samples added.

Observed/Inferred HCP Sentiment

☐ Positive
 ☒ Neutral
 ☐ Negative

Outcomes

Follow-up Actions

AI Suggested Follow-ups:

- Schedule follow-up meeting in 2 weeks
- Send OncoBoost Phase III PDF
- Add Dr. Sharma to advisory board invite list

AI Assistant
Log interaction via chat

Log interaction details here (e.g., "Met Dr. Smith, discussed Product X efficacy, positive sentiment, shared brochure") or ask for help.

Video URL: 📺 Instructions(Task 1).mp4

or use the URL below:

https://drive.google.com/file/d/1njXfNIWa465MfMYwY7YZvY_hGd3fyZX2/view?usp=drive_link

LangGraph AI Agent & Tools:

- Describe the role of the LangGraph agent in managing HCP interactions.
- Define a minimum of five (5) specific "tools" that this LangGraph agent would use for sales-related activities. Two tools must be:
 - **Log Interaction:** Detail how this tool captures interaction data (potentially using the LLM for summarization, entity extraction, etc.).
 - **Edit Interaction:** How this tool allows modification of logged data.

Output Deliverables

1. GitHub Submission

- Upload your frontend and backend code in one GitHub repository.
- Add a clear `README.md` explaining your project and how to run it.
- Share the GitHub repository link using the submission link at the end of this document.

2. Video Recording (10–15 minutes)

In your video, include:

- A walkthrough of your frontend (show how it works).
- A demo of all 5 LangGraph tools working properly.
- A simple explanation of your code and how your project is structured.
- A brief summary of what you understood from the task.



Submission Link

Submit both files separately using this Google Form:

<https://forms.gle/XdvLNBJkbdVDGADM8>